

National Forensic Sciences University
School of Cyber Security and Digital Forensics

Course Name: M.Sc. Cyber Security (Batch: 2025-27)

250103002004.

Semester - I

Subject Code: CTMSCS SII P1

Time: 11.00 AM-12.30 PM

Subject Name: Network Security

Exam: Mid Semester Examination (March-2026)

Date: 16-March-2026

Q1. Answer the following questions in short. (Attempt any 4)

[20 Marks]

- ✓1) Define different types of evidence and their importance in network forensics.
- 2) Explain following commands in detail: nmap 192.168.1.1 -sS and nmap 192.168.1.1 -sA
- 3) Describe different Timing and Performance options available in NMAP.
- ✓4) Create the cypher text for the message = "Sheistheattacker" with Key=5 with monoalphabetic Symmetric key cryptography.
- ✓5) Explain Combination Transposition Cipher in detail with example.
- ✓6) Explain the concept of Digital Signatures.

Q2. Answer the following questions in detail. (Attempt any 3)

[30 Marks]

✓1) Discuss Penetration Testing Life Cycle- Phases in detail.

2) Explain DES Algorithm with complete architecture.

✓3) Discuss RSA with suitable example. $\Rightarrow$ ans didn't come but all good.

✓4) Discuss OSCAR Methodology for network forensic analysis with suitable case study example. $\Rightarrow$ Case study not written

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M Sc Cyber Security, Semester – II

250103002004

CTMSCS SII P3: Mobile Security, Exam: Mid Semester Examination

Date: 18th March 2026

[Marks: 50]

Que – I Answer the following questions (Attempt any five)

- ✓1. Explain the different Android application components. Discuss the function of Activities, Services, Broadcast Receivers, and Content Providers with suitable examples
- ✓2. Discuss the exploitation of common Android application vulnerabilities such as insecure login, hardcoded data, insecure storage, input validation flaws, access control issues, content provider leakage, path traversal, and OWASP Mobile Top 10 risks.
- ✓3. Explain the reverse engineering process of android application and how does pen-tester use it for pen-testing?
4. Discuss Android Application audit using Drozer.
- ✓5. Discuss the Android Application static audit using MobSF.
- ✓6. Discuss Android Boot Process in detail.

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Course Name: M.Sc. Cyber Security (Batch: 2025-30)
Semester - II

Subject Code: CTMSCS SII P3
Subject Name: Malware Analysis
Exam: Mid Semester Examination (March - 2026)

Time: 1100-1230
Total Marks: 50
Date: 19-03-2026

Q1. Answer the following questions in short. (Attempt any 4) [20 Marks]

- ✓ 1) Explain any five types of malwares with its characteristics briefly.
- ✓ 2) How to create a YARA rule based malware signature? Explain with practical example.
- 3) Discuss any five dlls with its description of use.
- 4) Explain the disassembly and converts its C construct code from code-1.
- 5) Understand the following C code and create its disassembly from code-2.

Q2. Answer the following questions in detail. (Attempt any 3) [24 Marks]

- ✓ 1) How many seconds and minutes the program will sleep? Calculate it as per the given scenario and explain step by step.

```
.text:10001341 loc_10001341: ; CODE XREF: sub_10001074+10F↑j
.text:10001341 mov eax,off_10019010
.text:10001346 add eax, 0Eh
.text:10001349 push eax ; char*
.text:1000134A call ds:atoi
.text:10001350 mul eax, 960h = 38400
.text:10001356 pop ecx
.text:10001357 push eax ; dwmilliseconds
.text:10001358 call ds:sleep

.....
.data:10019010 off_10019010 db 'Class of MScCS25'; DATA XREF: sub_10001074:loc_10001341↑r
```

9x16 = 40
6x16 = 36
= 15
38400 = 10
10
10
10
10

- 2) Explain the assembly with its meaning as given below:

```
.text:0040148D loc_40148D: ; CODE XREF: _main+46↑j
.text:0040148D test eax, eax
.text:0040148F jnz loc_401813
.text:00401495 mov edi, ds:CreateFileA
.text:0040149B push eax ; hTemplateFile
.text:0040149C push eax ; dwFlagsAndAttributes
.text:0040149D push 1 ; dwCreationDisposition
.text:0040149F push eax ; lpSecurityAttributes
.text:004014A0 push 0 ; dwShareMode
.text:004014A2 push 40000000h ; dwDesiredAccess
.text:004014A7 push offset FileName ; "C:\\Windows\\System32\\Kerne132.dll"
.text:004014AC call edi ; CreateFileA
```

- 3) Explain CPU registers in detail. *segment registers*
- 4) Discuss all the malware analysis techniques and their merit and demerits with example.

Q3. Select the Correct answer from the options (Attempt all)

[06 Marks]

- 1) sigtool tool is used to create a CLAMAV signatures.
- 2) dll can be executed manually by rundll32.exe exe file with dll name, function name command.
- 3) strings in memory can be obtain by Strings tool.
- 4) Shortcut key to see all cross references of any function/window api in IDA is X.
- 5) Call type of graph is useful if analyst would like to know that a function calls what and how many windows api(s).
- 6) Unconditional jump is denoted in blue color in graph mode of IDA view.

```
.text:00101480 triangle proc near
.text:00101480 var_8 = dword ptr -8
.text:00101480 var_4 = dword ptr -4
.text:00101480 push    ebp
.text:00101481 mov     ebp, esp
.text:00101483 sub     esp, 8
.text:00101486 mov     [ebp+var_4], 1
.text:0010148D loc_10148D:
.text:0010148D mov     eax, [ebp+var_4]
.text:00101490 cmp     eax, 5
.text:00101493 jg      loc_1014C8
.text:00101495 mov     [ebp+var_8], 1
.text:0010149C loc_10149C:
.text:0010149C mov     eax, [ebp+var_8]
.text:0010149F cmp     eax, [ebp+var_4]
.text:001014A2 jg      loc_1014B6
.text:001014A4 push    offset aStar
.text:001014A9 call    _printf
.text:001014AE add     esp, 4
.text:001014B1 inc     [ebp+var_8]
.text:001014B4 jmp     loc_10149C
.text:001014B6 loc_1014B6:
.text:001014B6 push    offset aNewline
.text:001014BB call    _printf
.text:001014C0 add     esp, 4
.text:001014C3 inc     [ebp+var_4]
.text:001014C6 jmp     loc_10148D
.text:001014C8 loc_1014C8:
.text:001014C8 mov     esp, ebp
.text:001014CA pop     ebp
.text:001014CB retn
.text:001014CB triangle endp
```

Code-2

```
#include <stdio.h>
int main() {
    int number;
    printf("Enter a number: ");
    scanf("%d", &number);

    if (number > 0) {
        printf("The number is positive.\n");
    }
    else if (number < 0) {
        printf("The number is negative.\n");
    }
    else {
        printf("The number is zero.\n");
    }

    .....
    .....
}
```

← Code-1